

Color theory for screens, not paintings

Week 4, Lecture 1 · Design for Builders: Ship Beautiful Products as a Founder

Autumn 2026 · Founders, indie hackers, engineers

What we'll cover today

- **HSL and OKLCH:** the two color models you actually need
- **Perceptual uniformity:** why equal HSL steps look unequal on screen
- **Pure black and pure white:** why they belong in almost no UI
- **Accent vs neutral:** the two-ramp structure every product palette needs

Two color models worth knowing

HSL: the model you already use

Three channels: Hue (0-360°), Saturation (0-100%), Lightness (0-100%).

- Hue 0 = red, 120 = green, 240 = blue
- Saturation 0 = gray at any hue, 100 = most vivid
- Lightness 0 = always black, 100 = always white

CSS: `hsl(216, 80%, 52%)`, readable, editable, the default in design tools.

The HSL problem: equal steps, unequal brightness

Set three colors to the same HSL lightness, say, 50%:

- Yellow at `hsl(60, 100%, 50%)`
- Blue at `hsl(240, 100%, 50%)`
- Red at `hsl(0, 100%, 50%)`

Place them side by side. Yellow looks dramatically lighter than blue.

HSL lightness is not perceptual brightness. (Koopersmith & Miner, Stripe, 2019)

Why this matters for ramps

If you generate a ramp by incrementing HSL lightness by 10% per step, some steps will look far apart and others will look almost identical.

The ramp will look uneven even though the numbers are perfectly regular.

Fix: adjust individual steps by eye, or switch to OKLCH.

OKLCH: perceptual uniformity built in

Three channels: Lightness (L), Chroma (C), Hue (H).

- L is calibrated to human vision: equal steps in L look equal
- C is chroma (equivalent to saturation, but perceptually uniform)
- H is hue angle, similar to HSL hue

CSS: `oklch(0.6 0.12 240)`, available in Figma and Chrome DevTools.

Ramps built in OKLCH need less hand-tuning than ramps built in HSL.

HSL vs OKLCH: which to use

HSL

- Available in all tools today
- Familiar from CSS
- Manual step adjustment needed
- Fine for hand-tuned ramps

OKLCH

- Available in Figma (2024) and modern browsers
- Better for automated ramp generation
- Less manual correction needed
- Future-proof (CSS Color Level 4)

Why pure black and pure white are
wrong

Pure black: no room to move

`#000000` is the heaviest foreground possible. It creates harsh halation against light backgrounds.

More practically: if your darkest foreground is already `#000`, you have no darker state for pressed, active, or disabled variants.

The darkest value in a good neutral ramp is typically around 6-12% HSL lightness, not 0%.

Pure white: glare and no room to layer

`#ffffff` on screens (especially OLED) creates glare in low light.

More practically: if your lightest surface is `#fff`, a card on that surface has nowhere lighter to go. Card elevation requires a subtly different background.

The lightest value in a good neutral ramp is typically around 96-98% HSL lightness, not 100%.

Tinting your neutral ramp

A neutral ramp tinted toward your brand hue produces subtle harmony.

```
Pure gray:  hsl( 0, 0%, 97%)  -- clinical, flat  
Cool gray:  hsl(220, 9%, 97%)  -- slightly blue  
Warm gray:  hsl( 30, 6%, 97%)  -- slightly cream
```

Match the hue of your neutral to your accent color.

Low saturation (4-10%) is enough. More than 12% stops reading as neutral.

(Wathan & Schoger, Refactoring UI, 2019)

Accent vs neutral: the two-ramp structure

What a ramp is

A ramp is 9 shades of the same color family, labeled 100 (lightest) through 900 (darkest), with an optional 50 for near-white surfaces.

Each step has a clear purpose:

```
100-200: tinted surfaces, badge backgrounds  
300-400: borders, dividers, disabled states  
500:      base color (closest to the brand)  
600-700: text on light surfaces, interactive elements  
800-900: high-emphasis foreground on light surfaces
```

The accent ramp

Your accent is your brand color, the hue you use for buttons, links, and primary interactive elements.

Build 9 steps around your base:

- Steps 100-200: pastel backgrounds for alerts, badges
- Step 500: brand color
- Steps 600-700: interactive elements on light surfaces (usually darker than 500 for contrast)

Saturated colors need more correction than neutrals because perceptual brightness varies more steeply at high saturation.

The neutral ramp

Neutrals carry everything else: text, backgrounds, borders, dividers.

One neutral ramp is usually enough for a product. Use a single hue throughout (slightly blue, slightly warm, or pure) and keep saturation low and consistent.

```
Neutral/50:  hsl(220,  9%, 97%)  -- page background  
Neutral/500: hsl(220,  9%, 50%)  -- mid-gray  
Neutral/900: hsl(220,  9%, 11%)  -- primary foreground
```

The whole ramp is 9 values. That is enough for any product.

Building the ramp: the three-step process

Wathan & Schoger (Refactoring UI, 2019):

1. **Choose the base.** Usually step 500, but adjust if the brand color is very dark or light.
2. **Define the endpoints.** Step 100 = lightest useful value. Step 900 = darkest useful value.
Neither is `#fff` or `#000`.
3. **Fill the middle.** Work outward from the base. Adjust each stop until the visual progression looks even.

Trust your eyes over arithmetic.

Take-aways

1. Use HSL for manual ramp tuning; use OKLCH if you want automated steps to look perceptually even.
2. Pure black and pure white belong in almost no production palette. Near-extremes give you room for layering and interactive states.
3. Every product palette needs two ramps: a neutral (for text, surfaces, borders) and an accent (for interactive elements and brand color).
4. A ramp generator is a first draft. Check every stop visually and adjust until the progression looks even, not just numerically regular.

"When you actually grab the eyedropper and inspect those dark text colors, they're usually a very dark navy or brown, not black."

Adam Wathan & Steve Schoger, Refactoring UI (2019)

What's next

- **Lecture 2 (this week):** Tailwind-style 9-step ramps, WCAG contrast, semantic tokens, dark mode as a parallel system
- **Reading:** Week 4 reading, Color, palettes, contrast, and dark mode
- **Section:** Build your product's full palette as Figma variables

No assignment due this week.